Congenital Heart Disease — Surgery

The Big Picture

CHD = structural heart defect present at birth. Surgeon's job: know **which lesion**, **whether it needs surgery**, and **when** to operate (timing is everything in babies). Two broad surgery types: **palliative** (buy time, improve flow) and **reparative** (fix it — corrective).

Why It Happens

- **Genetic (10%)** — chromosomal (5-10%, e.g. Trisomy 21) + single mutant gene (3%).
- **Genetic–environmental (90%)** — multifactorial, the majority. Includes teratogens.
- **Teratogens** — Drugs (alcohol, amphetamines, anticonvulsants, lithium, retinoic acid, thalidomide), Infections (Rubella, Coxsackie), Maternal conditions (diabetes, SLE → heart block, PKU, old age).

How Common (and the shunt buckets)

- **Left→Right shunt (53%)** — the big group. VSD, ASD, PDA, AVSD. (Acyanotic.)
- **Right→Left shunt (11%)** — cyanotic. TOF (4.5%, the classic), tricuspid atresia, pulmonary atresia.
- **Admixture (15%)** — TGA (5%), single ventricle.
- **Obstructive (15%)** — Coarctation (9.5%), pulmonary stenosis.

Reading the Baby

- **Infant** — murmur, heart failure signs (poor feeding, low weight gain, tachypnea, sweating), cyanosis + hypoxic spells.
- **Child** — murmur, exercise intolerance, syncope, chest pain, clubbing.

Palliative Surgery (buy time)

- **Systemic–pulmonary shunt (Blalock-Taussig)** — for severe cyanosis / low pulmonary flow; grows a hypoplastic PA.
- **Pulmonary artery banding** — *too much* lung flow; protects vascular bed, controls CHF.
- **Atrial septectomy** — improves mixing (TGA, tricuspid atresia).

The Key Lesions & When to Fix

- **PDA** — Aorta↔LPA link. Significant: close after 1st month. HF/failure to thrive: any time. Severe pulmonary vascular disease: don't.
- **Coarctation** — narrow upper thoracic aorta. Operate if lumen ↓>50%, upper-body HTN, CHF, or with VSD.
- **ASD** — atrial hole, most common malformation. Fix if Qp/Qs >1.5–2.0; ideally age 1–5.
- **VSD** — large symptomatic → operate (before 3 mo if CHF). Small (Qp/Qs<1.5) → leave, but endocarditis risk.
- **TOF** — RV infundibulum underdeveloped. Diagnosis IS the indication. Total correction 3–24 mo (shunt first if symptomatic early).
- **TGA** — Aorta off RV, PA off LV. **Arterial switch within 1 month**. Beyond 30 days → atrial switch (Mustard/Senning).
- **Tricuspid atresia** — single-ventricle pathway: shunt → BCPS (6–12mo) → **Fontan** (12–24mo). PVR drives it.

- **HLHS** — staged: **Norwood** (1–30d) → BCPS/hemi-Fontan (6–12mo) → completion Fontan (18–24mo).

Last Resort

Cardiac transplant — poor prognosis <1yr survival, not fixable conventionally. Contraindicated if severe coexisting disease, active infection, or PVR >8 Wood units.

Walk-Away Points

- Timing is the exam answer — memorise the windows.
- **Arterial switch <1 month** for simple TGA.
- **Norwood** → **Fontan** staged path = HLHS & single ventricle.
- L→R = acyanotic & common; R→L (TOF) = cyanotic.
- PVR too high (>8 Wood) kills your surgical options.

Congenital Heart Disease — Revision Table

Lesion	Shunt/Type	Key fact	Surgery & timing
PDA	L→R (17%)	Aorta↔proximal LPA	Close after 1st mo; any time if HF; CI if severe PVD
ASD	L→R (16.5%)	Most common malformation	Repair if Qp/Qs>1.5–2; age 1–5
VSD	L→R (13%)	Hole between ventricles	Large+CHF: <3mo; small: observe (IE risk)
AVSD	L→R (3.5%)	Endocardial cushion defect	Partial 1–2yr; complete 3–6mo
Coarctation	Obstructive (9.5%)	Narrow upper thoracic aorta	Lumen↓>50%, HTN, CHF, +VSD
Pulmonary stenosis	Obstructive	RVOT obstruction	Sx + gradient >50mmHg; balloon valvotomy
TOF	R→L (4.5%)	RV infundibulum underdev., cyanotic	Dx = indication; total correction 3–24mo
TGA	Admixture (5%)	Aorta off RV, PA off LV	**Arterial switch <1mo**; later → atrial switch
Tricuspid atresia	R→L (3%)	Univentricular AV connection	Shunt → BCPS 6–12mo → **Fontan** 12–24mo
HLHS	(0.9%)	Hypoplastic left structures	**Norwood** 1–30d → BCPS → Fontan
Ebstein	Valvular	TV leaflets displaced into RV	Repair + ASD closure if symptomatic; WPW ablation

Palliative op	Purpose	Used in
Blalock-Taussig shunt	↑ pulmonary flow	Severe cyanosis, low lung flow
PA banding	↓ pulmonary flow	Large/multiple VSD, CHF
Atrial septectomy	↑ mixing	TGA, tricuspid/pulmonary atresia

Cause group	Examples
Genetic (10%)	Chromosomal (Trisomy 21), single gene
Teratogen drugs	Alcohol, thalidomide, retinoic acid, anticonvulsants, lithium
Teratogen infection	Rubella, Coxsackie B
Maternal	Diabetes (VSD/TGA), SLE (heart block), PKU, old age

Transplant	Detail
Indication	Poor prognosis <1yr, not treatable conventionally
Contraindication	Severe coexisting disease, active infection, **PVR >8 Wood units**

Bold = high-yield.